TEXORA · LECTURE SHEET

Electrical Properties

Fibres as reluctant conductors: volume, surface and mass specific resistance, conduction through sorbed water, the overwhelming influence of relative humidity, dielectric behaviour — the properties that decide whether charge leaks away quietly or accumulates into the static problems of the next topic.

TE-3209 · TEXTILE PHYSICS · TOPIC 9 · LEVEL 3 TERM 2

1. Resistance in a textile geometry

Dry fibres are insulators, but textile forms complicate the bookkeeping: currents travel *through* fibres (volume conduction) and *along* them (surface conduction, usually dominant when any moisture film exists). Three measures coexist: **volume resistivity** ρ (Ω·cm) from R = ρl/A on a defined specimen; **surface resistivity** (Ω per square) for fabrics; and the textile trade's practical **mass specific resistance**, the resistance between electrodes applied to a standard mass/length arrangement of fibres, quoted because real fibre assemblies cannot be machined into neat bars. Orders of magnitude tell the story (at 65% RH, indicative): hydrophilic fibres sit near 10^{8}–10^{10} Ω·cm; the same fibres oven-dry, and hydrophobic synthetics even when conditioned, reach 10^{13}–10^{16} Ω·cm — a span wider than the gap between copper and glass in ordinary engineering.

2. What carries the current — and what humidity does

Conduction in fibres is chiefly **ionic**: sorbed water plus trace salts and ionisable groups provide mobile charge carriers in the amorphous regions. Hence the master rule of this topic: resistance is a function of regain, and therefore of relative humidity, falling roughly exponentially as RH rises — a useful rule of thumb is an order-of-magnitude fall for about each 10% RH rise across the working range for cellulosic fibres. Crystallinity excludes carriers (conduction is an amorphous-region property, Topic 1), temperature assists it, and surface finishes (antistatic agents are hygroscopic or ionic coatings) manipulate it directly.

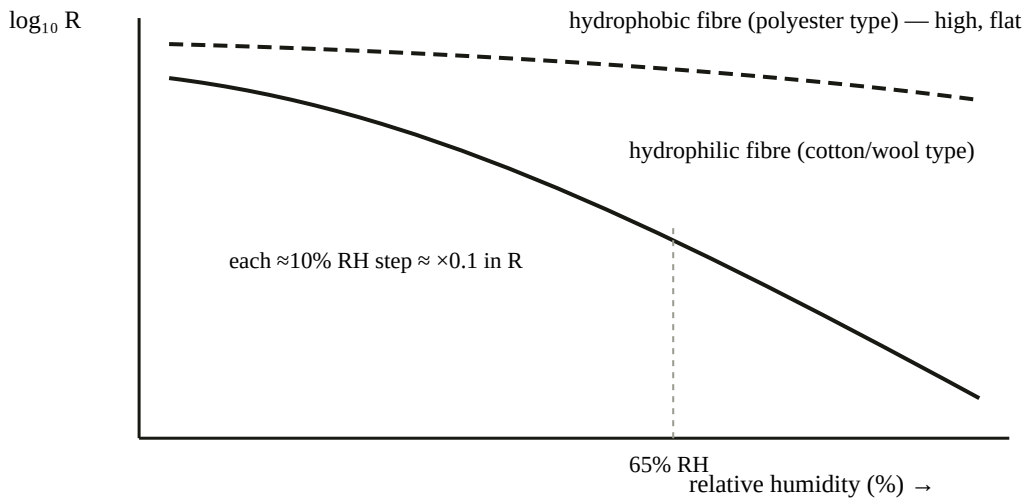


Figure 1 — Resistance versus relative humidity (schematic, log scale): hydrophilic fibres plunge as RH rises; hydrophobic fibres stay high and nearly flat — the root of their static reputation.

WORKED EXAMPLE — THE HUMIDITY LEVER IN NUMBERS

A cotton web measures R ≈ 10^{10} Ω between standard electrodes at 45% RH. Applying the rule of thumb (×0.1 per +10% RH): at 65% RH, R ≈ $10^{10} \times 10^{-2}$ = **10^{8} Ω**; at 85% RH, ≈ **10^{6} Ω**. Two practical morals: a static complaint that vanishes on humid days is this arithmetic announcing itself, and conditioning a laboratory to 65% RH is not pedantry — electrical (and mechanical) comparisons across days are meaningless without it.

3. Dielectric behaviour

In an alternating field a fibre polarises: the **dielectric constant** (relative permittivity) measures stored charge, and dielectric loss measures the lag dissipated as heat. Dry fibres sit at ε_r ≈ 2–5, but sorbed water (ε_r ≈ 80) drags the value steeply upward with regain — so dielectric measurement is a sensitive **moisture meter**: capacitance-based regain and moisture-content gauges on dryers and size boxes are this property at work. The same loss mechanism, driven hard, is **radio-frequency (RF) drying**: wet regions absorb the field preferentially and dry themselves level — self-levelling moisture removal that hot air cannot imitate.

4. Key points

- Textile conduction is ionic, through sorbed water in amorphous regions; dry and hydrophobic fibres reach 10^{13}–10^{16} Ω·cm territory.
- Resistance plunges with RH — roughly ×0.1 per +10% RH for cellulosics (10^{10} Ω at 45% → 10^{6} at 85% in the illustration).

- Surface conduction usually dominates in real assemblies; specify geometry and atmosphere or the number is meaningless.
- Dielectric constant tracks moisture — the principle of capacitance regain meters and of self-levelling RF drying.
- These resistances set the leak rate that decides static accumulation (Topic 10): generation is friction's business, escape is resistivity's.

5. Practice

Answer in your own words; brief model answers follow.

1. A viscose fabric measures 10^9 Ω at 55% RH. Estimate its resistance at 75% RH using the sheet's rule of thumb, and state why the estimate is safer as an order of magnitude than as a value.
2. Why does a polyester fabric, taken from the same 65% RH room as a cotton one, still hold charge while the cotton does not?
3. Explain the self-levelling action of RF drying in dielectric terms.
4. A capacitance moisture meter reads differently on two fabrics of equal true regain but different salt content. Why?
5. Why must electrical tests on textiles specify the electrode arrangement and conditioning so rigidly, compared with a metal resistivity test?
6. An antistatic finish claims to "conduct static away". Translate the claim into the quantities of this topic and name its weakness in very dry air.

Answer key

1. Two +10% steps → $\times 10^{-2}$ → ≈ 10^7 Ω, as an order of magnitude. The rule is an average slope of a curved, hysteretic relation (Topic 4); individual fibres, finishes and history move the constant — the exponent is informative, the mantissa is not.
2. Polyester's regain at 65% RH is ≈ 0.4% against cotton's ≈ 7–8%: it lacks the sorbed-water ion paths, so its resistance remains orders higher and charge generated on it leaks away far more slowly — same room, different internal weather.
3. Dielectric loss rises steeply with local moisture content: wetter regions absorb more RF power and heat/dry faster than dry ones, so the process automatically concentrates energy where water remains — negative feedback toward uniform dryness.
4. Ionic salts raise conduction and dielectric loss independently of water content; the meter responds to the electrical state, not to water alone — contamination masquerades as moisture. Calibration is per-material and per-process-chemistry.

5. Because the specimen is an anisotropic, porous, humidity-sensitive network: current paths depend on fibre direction, pressure, contact area and sorption history. A metal bar is homogeneous and stable; a fibre assembly is none of those, so geometry and atmosphere are part of the measurand.
6. The finish lowers surface resistivity (hygroscopic/ionic layer) so generated charge leaks to earth or spreads and neutralises instead of accumulating. In very dry air its own water film thins and its resistance climbs back — humidity-dependent antistats fail exactly when static is worst, which is why durable/integral solutions (conductive fibres, Topic 10) exist.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib